



8735 - Measuring sizes of star clusters in nearby galaxies using PHANGS

Cycle: 4, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Oleg Y. Gnedin (PI)	University of Michigan
Dr. Janice Lee (CoI)	Space Telescope Science Institute
Dr. David Thilker (CoI)	The Johns Hopkins University
Dr. Daniel Maschmann (CoI)	Space Telescope Science Institute
Dr. Thomas Williams (CoI) (ESA Member)	University of Oxford
Dr. Daniel Dale (CoI)	University of Wyoming
Prof. Frank Bigiel (CoI) (ESA Member)	Universitat Bonn, Argelander Institute for Astronomy
Dr. Hwihyun Kim (CoI)	International Gemini Observatory/NSF NOIRLab

OBSERVATIONS

ABSTRACT

We propose to measure the radii of 36,000 compact star clusters in 38 nearby galaxies, the largest such sample to-date, using new JWST NIRCам imaging. The study outlined in this AR proposal will increase the number of clusters with published radii by a factor of four, and double the number of galaxies with such measurements. Compared to previous best radius measurements with HST, JWST imaging offers higher spatial resolution, less extinction, and better SED coverage. We will adapt a publicly-available custom pipeline that has been demonstrated to accurately and robustly measure the radii of star clusters in HST imaging. With this uniformly measured sample we will obtain the distribution of cluster radii and internal densities, determine the cluster mass-radius relation, and investigate the variation of cluster properties with age across diverse galactic environments. These measurements will enable an unprecedented examination of the formation and long-term evolution of star clusters.

OBSERVING DESCRIPTION